

Akash Punnam

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SKILLS

- **Programming Languages:** Python, SQL, Scala
- **Database Management Systems (DBMS):** MySQL, PostgreSQL, Oracle, SQL Server, MongoDB, Cassandra, DBT
- **Data Warehousing:** Amazon Redshift, Google BigQuery, Snowflake, Apache Hive
- **ETL Tools:** Apache Spark, Apache Airflow, Talend, Informatica, Apache NiFi
- **Big Data Technologies:** Apache Hadoop, Apache Kafka, Apache HBase, Apache Flink, Apache Storm
- **Cloud Platforms:** AWS, Azure, GCP, including services like Amazon S3, Azure Data Lake Storage
- **Version Control Systems:** Git, GitHub
- **Data Visualization:** Tableau, Power BI
- **Machine Learning/AI:** TensorFlow, PyTorch
- **Operating System:** Linux/Unix, Windows, macOS
- **Containerization and Orchestration:** Docker, Kubernetes
- **Tools & IDE:** Git, IntelliJ, Visual Studio Code, Jupyter Notebook, PyCharm
- **Hadoop Ecosystem:** HDFS, YARN, MapReduce
- **Monitoring and Logging:** Prometheus, Grafana, ELK stack
- **Analytical Tools:** SAS, Microsoft Power BI

EDUCATION

Master of Engineering in Computer science

University of Cincinnati, Cincinnati, Ohio

August 2023 – May 2025

Cumulative GPA: 3.5/4

EXPERIENCE

DATA ENGINEER

July 2019 – June 2023

Sunrisemithra Global Technologies, Khammam, India

- Executed a thorough review process for software features that led to findings addressing three critical causes of system downtime; boosted application stability by 40% through targeted enhancements within the development cycle.
- Orchestrated comprehensive testing protocols surrounding new views created in the system ensuring reliability before deployment which resulted in zero post-launch defects reported during initial usage phases within first month.
- Led initiatives focusing on streamlined database interactions via PL/SQL Stored Procedures, achieving an average decrease of 25 hours monthly spent troubleshooting issues related to inefficient queries or processes.
- Architected an innovative information catalog framework employing cutting-edge technologies such as Elasticsearch and AWS services; achieved efficient real-time data analysis while reducing query response times considerably for over 100 datasets.
- Spearheaded enhancements to data querying capabilities using well-structured Materialized Views combined with sophisticated Indexing approaches; these efforts resolved three recurring bottlenecks affecting system throughput and user satisfaction levels.
- Orchestrated comprehensive data analysis workflows through the application of powerful Scala functions combined with essential Spark Transformation techniques, achieving improved accuracy for datasets analyzed weekly exceeding two terabytes.
- Orchestrated comprehensive testing procedures using custom-built PySpark code which identified inconsistencies in over 10TB of archived files across both HDFS and AWS environments; findings directly supported infrastructure improvements preventing potential errors.
- Implemented scalable ETL frameworks designed specifically for transferring multi-terabyte datasets seamlessly into Amazon's analytical platform (AWS Redshift) while ensuring minimal downtime during migrations performed weekly over three months.
- Automated data ingestion from S3 (CSV, JSON, Parquet) into Snowflake, DynamoDB, and AWS S3 using AWS Glue and crawlers.
- Established seamless integration points between dimension and fact tables during ODS development in Snowflake; this resulted in an impressive increase of over three-fold reduction time when running complex queries against large datasets.
- Implemented rigorous unit tests on datasets transferred between Redshift and Snowflake, ensuring optimal data consistency; findings led to the resolution of three critical issues affecting database performance.

- Streamlined Continuous Delivery processes by leveraging Docker and GitHub, achieving a deployment time reduction of 40%, allowing for more frequent updates to application features without service interruptions.
- Developed robust DAG orchestrations within Apache Airflow to streamline ETL processes, leading to significant enhancements in workflow reliability and contributing to a substantial reduction of manual intervention by over 50 hours per month.
- Optimized data aggregation workflows through the implementation of Spark-SQL in Databricks, resulting in a significant 30% reduction in processing time for extracting critical insights on customer behavior.
- Leveraged knowledge of user requirements as the most junior member on the analytics team by developing intuitive tableau reports reflecting core business needs—resulting in enhanced cross-department communication regarding ongoing projects.